

LAN (LOCAL AREA NETWORK)

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher security.



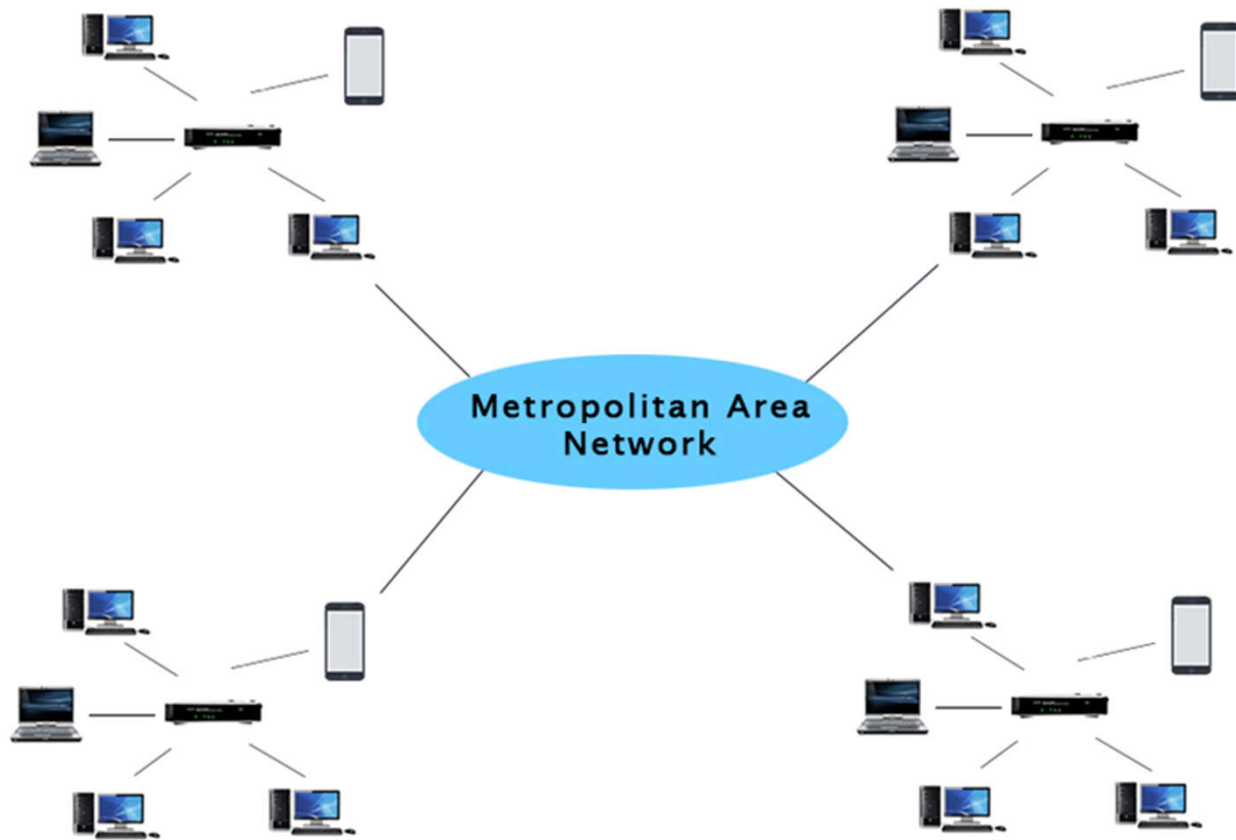
PAN(Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- **Thomas Zimmerman** was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network covers an area of **30 feet**.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.



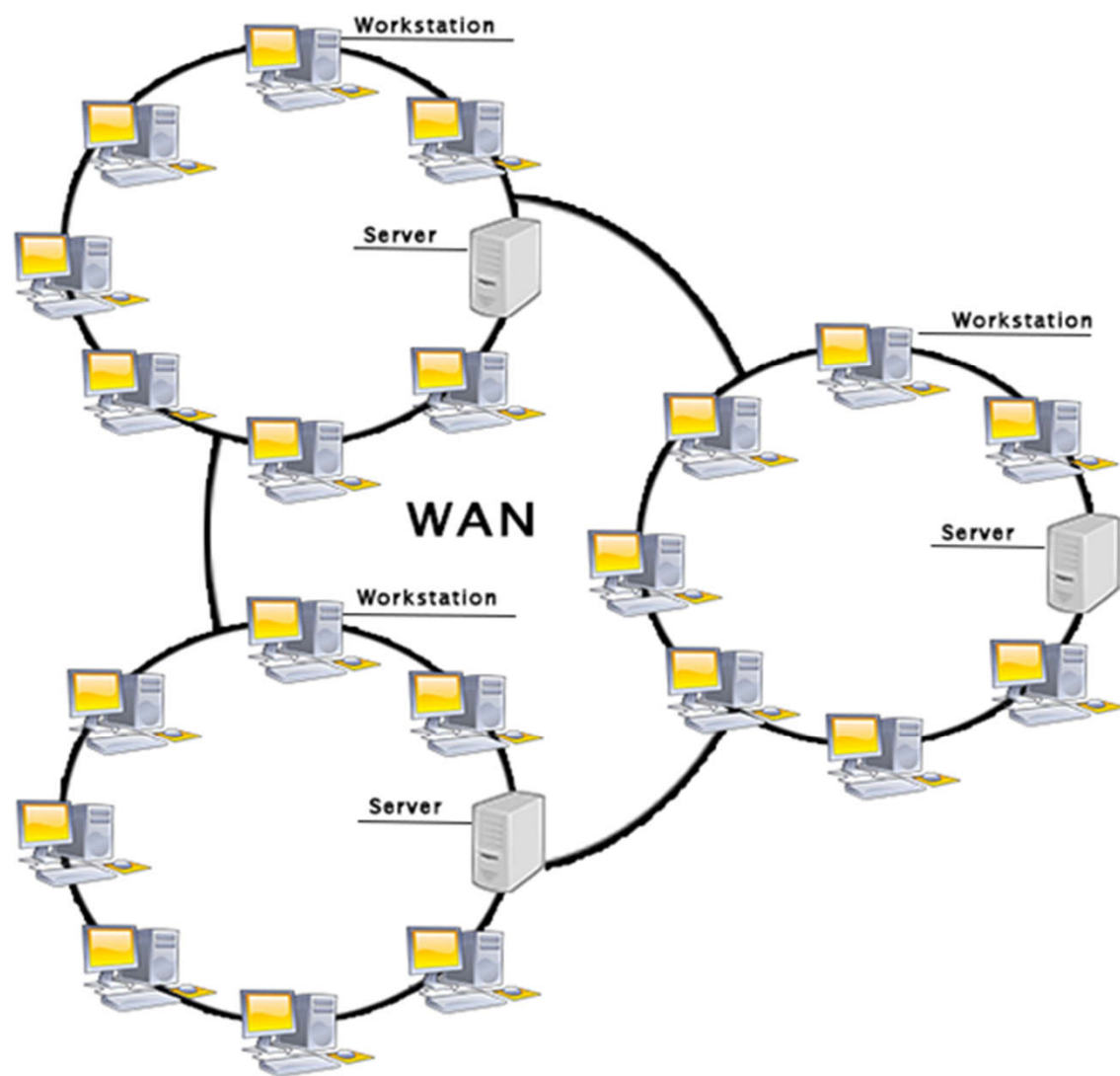
MAN(Metropolitan Area Network)

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, ADSL, etc.
- It has a higher range than Local Area Network(LAN).



- Uses Of Metropolitan Area Network:
- MAN is used in communication between the banks in a city.
- It can be used in an Airline Reservation.
- It can be used in a college within a city.
- It can also be used for communication in the military.
-

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.



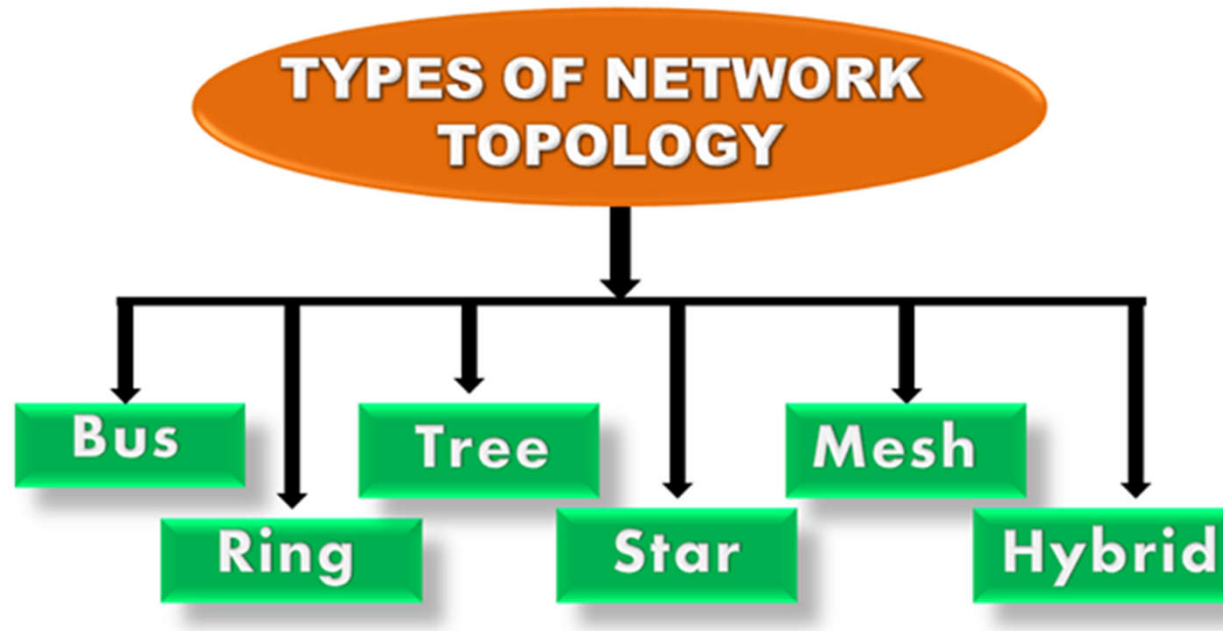
- Examples Of Wide Area Network:
- **Mobile Broadband:** A 4G network is widely used across a region or country.
- **Last mile:** A telecom company is used to provide the internet services to the customers in hundreds of cities by connecting their home with fiber.
- **Private network:** A bank provides a private network that connects the 44 offices. This network is made by using the telephone leased line provided by the telecom company.
- Advantages Of Wide Area Network:
- Following are the advantages of the Wide Area Network:
- **Geographical area:** A Wide Area Network provides a large geographical area. Suppose if the branch of our office is in a different city then we can connect with them through WAN. The internet provides a leased line through which we can connect with another branch.
- **Centralized data:** In case of WAN network, data is centralized. Therefore, we do not need to buy the emails, files or back up servers.
- **Get updated files:** Software companies work on the live server. Therefore, the programmers get the updated files within seconds.
- **Exchange messages:** In a WAN network, messages are transmitted fast. The web application like Facebook, Whatsapp, Skype allows you to communicate with friends.
- **Sharing of software and resources:** In WAN network, we can share the software and other resources like a hard drive, RAM.
- **Global business:** We can do the business over the internet globally.
- **High bandwidth:** If we use the leased lines for our company then this gives the high bandwidth. The high bandwidth increases the data transfer rate which in turn increases the productivity of our company.
-

- Disadvantages of Wide Area Network:
- The following are the disadvantages of the Wide Area Network:
- **Security issue:** A WAN network has more security issues as compared to LAN and MAN network as all the technologies are combined together that creates the security problem.
- **Needs Firewall & antivirus software:** The data is transferred on the internet which can be changed or hacked by the hackers, so the firewall needs to be used. Some people can inject the virus in our system so antivirus is needed to protect from such a virus.
- **High Setup cost:** An installation cost of the WAN network is high as it involves the purchasing of routers, switches.
- **Troubleshooting problems:** It covers a large area so fixing the problem is difficult.
- Internet is a vast network that connects computers all over the world.

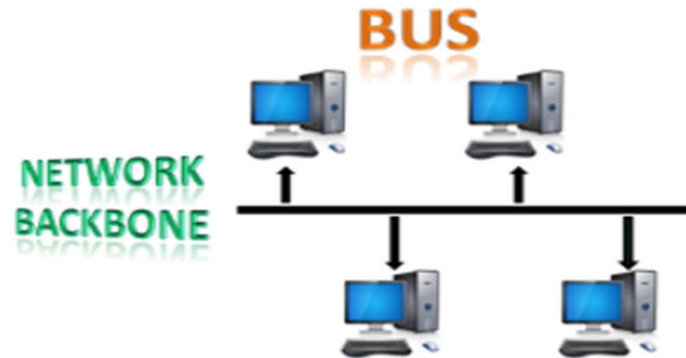
Topology

Topology defines the structure of the network of how all the components are interconnected to each other. There are two types of topology: physical and logical topology.

Physical topology is the geometric representation of all the nodes in a network.



Bus Topology



The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.

Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable. When a node wants to send a message over the network, it puts a message over the network. All the stations available in the network will receive the message whether it has been addressed or not.

The bus topology is mainly used in 802.3 (ethernet) and 802.4 standard networks.

The configuration of a bus topology is quite simpler as compared to other topologies.

The backbone cable is considered as a "**single lane**" through which the message is broadcast to all the stations.

The most common access method of the bus topologies is **CSMA** (Carrier Sense Multiple Access).

CSMA: It is a media access control used to control the data flow so that data integrity is maintained, i.e., the packets do not get lost. There are two alternative ways of handling the problems that occur when two nodes send the messages simultaneously.

CSMA CD: CSMA CD (**Collision detection**) is an access method used to detect the collision. Once the collision is detected, the sender will stop transmitting the data. Therefore, it works on "**recovery after the collision**".

CSMA CA: CSMA CA (**Collision Avoidance**) is an access method used to avoid the collision by checking whether the transmission media is busy or not. If busy, then the sender waits until the media becomes idle. This technique effectively reduces the possibility of the collision. It does not work on "recovery after the collision".

Advantages of Bus topology:

Low-cost cable: In bus topology, nodes are directly connected to the cable without passing through a hub. Therefore, the initial cost of installation is low.

Moderate data speeds: Coaxial or twisted pair cables are mainly used in bus-based networks that support upto 10 Mbps.

Familiar technology: Bus topology is a familiar technology as the installation and troubleshooting techniques are well known, and hardware components are easily available.

Limited failure: A failure in one node will not have any effect on other nodes.

Disadvantages of Bus topology:

Extensive cabling: A bus topology is quite simpler, but still it requires a lot of cabling.

Difficult troubleshooting: It requires specialized test equipment to determine the cable faults. If any fault occurs in the cable, then it would disrupt the communication for all the nodes.

Signal interference: If two nodes send the messages simultaneously, then the signals of both the nodes collide with each other.

Reconfiguration difficult: Adding new devices to the network would slow down the network.

Attenuation: Attenuation is a loss of signal leads to communication issues. Repeaters are used to regenerate the signal.

Ring Topology



Ring topology is like a bus topology, but with connected ends.

The node that receives the message from the previous computer will retransmit to the next node.

The data flows in one direction, i.e., it is unidirectional.

The data flows in a single loop continuously known as an endless loop.

It has no terminated ends, i.e., each node is connected to other node and having no termination point.

The data in a ring topology flow in a clockwise direction.

The most common access method of the ring topology is **token passing**.

Token passing: It is a network access method in which token is passed from one node to another node.

Token: It is a frame that circulates around the network

Working of Token passing

A token moves around the network, and it is passed from computer to computer until it reaches the destination. The sender modifies the token by putting the address along with the data.

The data is passed from one device to another device until the destination address matches. Once the token received by the destination device, then it sends the acknowledgment to the sender.

In a ring topology, a token is used as a carrier.

Advantages of Ring topology:

Network Management: Faulty devices can be removed from the network without bringing the network down.

Product availability: Many hardware and software tools for network operation and monitoring are available.

Cost: Twisted pair cabling is inexpensive and easily available. Therefore, the installation cost is very low.

Reliable: It is a more reliable network because the communication system is not dependent on the single host computer.

Advantages of Ring topology:

Network Management: Faulty devices can be removed from the network without bringing the network down.

Product availability: Many hardware and software tools for network operation and monitoring are available.

Cost: Twisted pair cabling is inexpensive and easily available. Therefore, the installation cost is very low.

Reliable: It is a more reliable network because the communication system is not dependent on the single host computer.

Disadvantages of Ring topology:

Difficult troubleshooting: It requires specialized test equipment to determine the cable faults. If any fault occurs in the cable, then it would disrupt the communication for all the nodes.

Failure: The breakdown in one station leads to the failure of the overall network.

Reconfiguration difficult: Adding new devices to the network would slow down the network.

Delay: Communication delay is directly proportional to the number of nodes. Adding new devices increases the communication delay.

Star Topology



Star topology is an arrangement of the network in which every node is connected to the central hub, switch or a central computer.

The central computer is known as a **server**, and the peripheral devices attached to the server are known as **clients**. Coaxial cable or RJ-45 cables are used to connect the computers.

Hubs or Switches are mainly used as connection devices in a physical star topology.

Star topology is the most popular topology in network implementation.

Advantages of Star topology

Efficient troubleshooting: Troubleshooting is quite efficient in a star topology as compared to bus topology. In a bus topology, the manager has to inspect the kilometers of cable. In a star topology, all the stations are connected to the centralized network. Therefore, the network administrator has to go to the single station to troubleshoot the problem.

Network control: Complex network control features can be easily implemented in the star topology. Any changes made in the star topology are automatically accommodated.

Limited failure: As each station is connected to the central hub with its own cable, therefore failure in one cable will not affect the entire network.

Familiar technology: Star topology is a familiar technology as its tools are cost-effective.

Easily expandable: It is easily expandable as new stations can be added to the open ports on the hub.

Cost effective: Star topology networks are cost-effective as it uses inexpensive coaxial cable.

High data speeds: It supports a bandwidth of approx 100Mbps. Ethernet 100BaseT is one of the most popular Star topology networks.

Disadvantages of Star topology

A Central point of failure: If the central hub or switch goes down, then all the connected nodes will not be able to communicate with each other.

Cable: Sometimes cable routing becomes difficult when a significant amount of routing is required.

Tree topology



- Tree topology combines the characteristics of bus topology and star topology.
- A tree topology is a type of structure in which all the computers are connected with each other in hierarchical fashion.
- The top-most node in tree topology is known as a root node, and all other nodes are the descendants of the root node.
- There is only one path exists between two nodes for the data transmission. Thus, it forms a parent-child hierarchy.

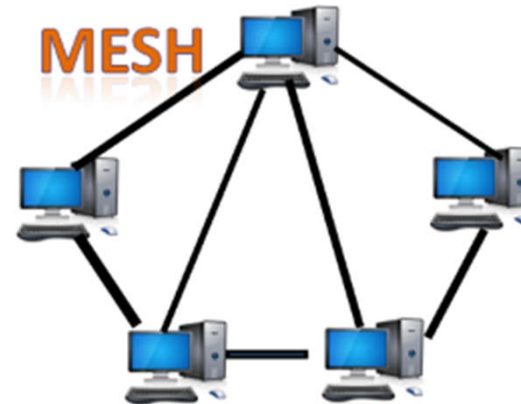
Advantages of Tree topology

- **Support for broadband transmission:** Tree topology is mainly used to provide broadband transmission, i.e., signals are sent over long distances without being attenuated.
- **Easily expandable:** We can add the new device to the existing network. Therefore, we can say that tree topology is easily expandable.
- **Easily manageable:** In tree topology, the whole network is divided into segments known as star networks which can be easily managed and maintained.
- **Error detection:** Error detection and error correction are very easy in a tree topology.
- **Limited failure:** The breakdown in one station does not affect the entire network.
- **Point-to-point wiring:** It has point-to-point wiring for individual segments.

Disadvantages of Tree topology

- **Difficult troubleshooting:** If any fault occurs in the node, then it becomes difficult to troubleshoot the problem.
- **High cost:** Devices required for broadband transmission are very costly.
- **Failure:** A tree topology mainly relies on main bus cable and failure in main bus cable will damage the overall network.
- **Reconfiguration difficult:** If new devices are added, then it becomes difficult to reconfigure.

Mesh topology



Mesh technology is an arrangement of the network in which computers are interconnected with each other through various redundant connections.

There are multiple paths from one computer to another computer.

It does not contain the switch, hub or any central computer which acts as a central point of communication.

The Internet is an example of the mesh topology.

Mesh topology is mainly used for WAN implementations where communication failures are a critical concern.

Mesh topology is mainly used for wireless networks.

Mesh topology can be formed by using the formula:

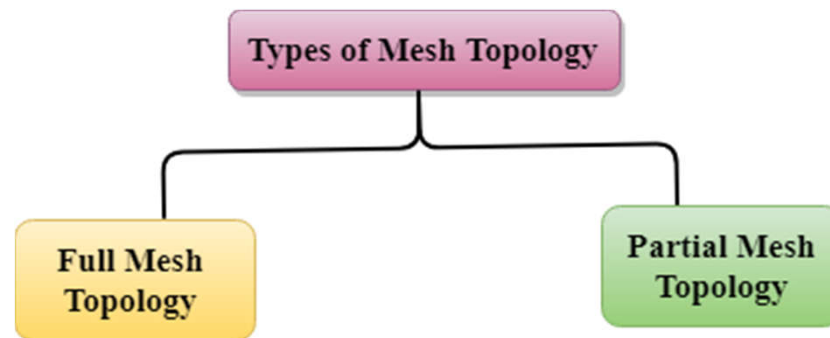
Number of cables = $(n*(n-1))/2$;

Where n is the number of nodes that represents the network.

Mesh topology is divided into two categories:

Fully connected mesh topology

Partially connected mesh topology



Full Mesh Topology: In a full mesh topology, each computer is connected to all the computers available in the network.

Partial Mesh Topology: In a partial mesh topology, not all but certain computers are connected to those computers with which they communicate frequently.

Advantages of Mesh topology:

Reliable: The mesh topology networks are very reliable as if any link breakdown will not affect the communication between connected computers.

Fast Communication: Communication is very fast between the nodes.

Easier Reconfiguration: Adding new devices would not disrupt the communication between other devices.

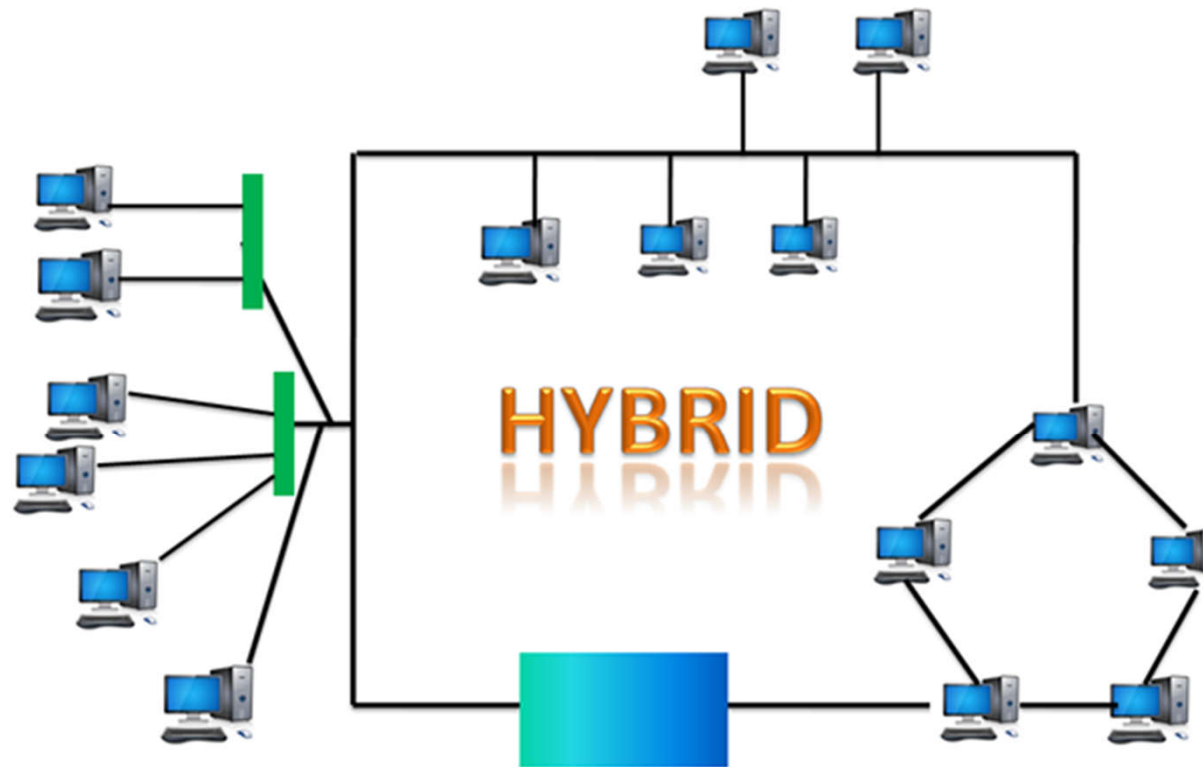
Disadvantages of Mesh topology

Cost: A mesh topology contains a large number of connected devices such as a router and more transmission media than other topologies.

Management: Mesh topology networks are very large and very difficult to maintain and manage. If the network is not monitored carefully, then the communication link failure goes undetected.

Efficiency: In this topology, redundant connections are high that reduces the efficiency of the network.

Hybrid Topology



The combination of various different topologies is known as **Hybrid topology**.

A Hybrid topology is a connection between different links and nodes to transfer the data.

When two or more different topologies are combined together is termed as Hybrid topology and if similar topologies are connected with each other will not result in Hybrid topology. For example, if there exist a ring topology in one branch of ICICI bank and bus topology in another branch of ICICI bank, connecting these two topologies will result in Hybrid topology.

Advantages of Hybrid Topology

Reliable: If a fault occurs in any part of the network will not affect the functioning of the rest of the network.

Scalable: Size of the network can be easily expanded by adding new devices without affecting the functionality of the existing network.

Flexible: This topology is very flexible as it can be designed according to the requirements of the organization.

Effective: Hybrid topology is very effective as it can be designed in such a way that the strength of the network is maximized and weakness of the network is minimized.

Disadvantages of Hybrid topology

Complex design: The major drawback of the Hybrid topology is the design of the Hybrid network. It is very difficult to design the architecture of the Hybrid network.

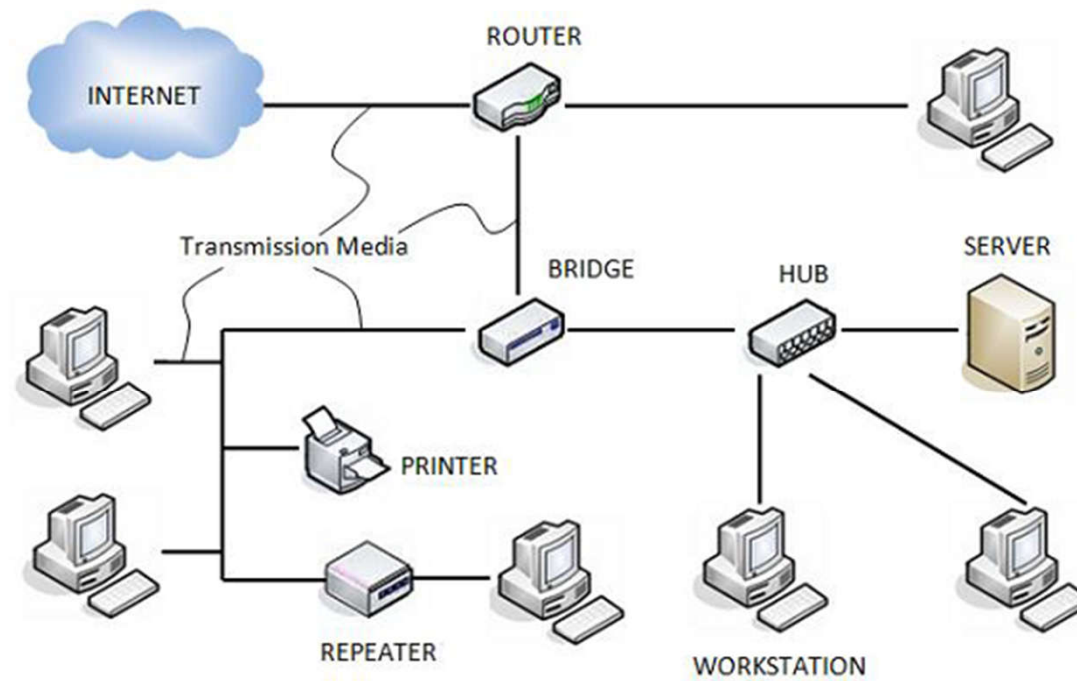
Costly Hub: The Hubs used in the Hybrid topology are very expensive as these hubs are different from usual Hubs used in other topologies.

Costly infrastructure: The infrastructure cost is very high as a hybrid network requires a lot of cabling, network devices, etc.

- Computer networks components comprise both physical parts as well as the software required for installing computer networks, both at organizations and at home. The hardware components are the server, client, peer, transmission medium, and connecting devices. The software components are operating system and protocols.

- Computer networks components comprise both physical parts as well as the software required for installing computer networks, both at organizations and at home. The hardware components are the server, client, peer (two devices having same operations), transmission medium, and connecting devices. The software components are operating system and protocols.

The following figure shows a network along with its components –



COMPUTER NETWORK COMPONENTS

- **Hardware Components**
- **Servers** – Servers are high-configuration computers that manage the resources of the network. The network operating system is typically installed in the server and so they give user accesses to the network resources. Servers can be of various kinds: file servers, database servers, print servers etc.
- **Clients** – Clients are computers that request and receive service from the servers to access and use the network resources.
- **Peers** – Peers are computers that provide as well as receive services from other peers in a workgroup network.
- **Transmission Media** – Transmission media are the channels through which data is transferred from one device to another in a network. Transmission media may be guided media like coaxial cable, fibre optic cables etc; or maybe unguided media like microwaves, infra-red waves etc.

The basic computer hardware components that are needed to set up a network are as follows –

Network Cables

Network cables are the transmission media to transfer data from one device to another.

A commonly used network cable is category 5 cable with RJ – 45 connector, as shown in the image below.



- **Connecting Devices** – Connecting devices act as middleware between networks or computers, by binding the network media together. Some of the common connecting devices are:
 - f. Routers
 - c. Bridges
 - b. Hubs
 - a. Repeaters
 - e. Gateways
 - d. Switches

Repeaters, Hubs, and Switches

Repeaters, hubs and switches connect network devices together so that they can function as a single segment.

A repeater receives a signal and regenerates it before re-transmitting so that it can travel longer distances (2 ports).

A hub is a multiport repeater having several input/output ports, so that input at any port is available at every other port.

A switch receives data from a port, uses packet switching (An **Ethernet** connection is faster, as it has a higher bandwidth) to resolve the destination device and then forwards the data to the particular destination, rather than broadcasting it as a hub.

A switch is a multiport bridge .



REPEATER



HUB



SWITCH

Bridges

A bridge connects two separate Ethernet network segments. It forwards packets from the source network to the destined network. It has a single input and single output port, thus making it a 2 port device.



Gateways

A gateway connects entirely different networks that work upon different protocols. It is the entry and the exit point of a network and controls access to other networks.

A gateway, as the name suggests, is a passage to connect two networks together that may work upon different networking models. They basically work as the messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switch or router.



Network Interface Cards

NIC is a component of the computer to connect it to a network. Network cards are of two types: Internal network cards and external network cards.



Modem

A modem is a hardware device that allows the computer to connect to the internet over the existing telephone line.

A modem is not integrated with the motherboard rather than it is installed on the PCI slot found on the motherboard.

It stands for Modulator/Demodulator. It converts the digital data into an analog(modulation) signal over the telephone lines.

The difference between a modem and a router is that **a modem connects to the internet, and a router connects devices to Wi-Fi.**

Cable Modem



ComputerHope.com

Routers

A router is a connecting device that transfers data packets between different computer networks. Typically, they are used to connect a PC or an organization's LAN to a broadband internet connection. They contain RJ-45 ports so that computers and other devices can connect with them using network cables. Routers normally connect LANs and WANs together



Software Components

Networking Operating System – Network Operating Systems is typically installed in the server and facilitate workstations in a network to share files, database, applications, printers etc.

Protocol Suite – A protocol is a rule or guideline followed by each computer for data communication. Protocol suite is a set of related protocols that are laid down for computer networks. The two popular protocol suites are

–

- a. OSI Model (Open System Interconnections)
- b. TCP / IP Model

Computer Network Models

A communication subsystem is a complex piece of Hardware and software. Early attempts for implementing the software for such subsystems were based on a single, complex, unstructured program with many interacting components. The resultant software was very difficult to test and modify. To overcome such problem, the ISO has developed a layered approach. In a layered approach, networking concept is divided into several layers, and each layer is assigned a particular task. Therefore, we can say that networking tasks depend upon the layers.

Layered Architecture

The main aim of the layered architecture is to divide the design into small pieces.

Each lower layer adds its services to the higher layer to provide a full set of services to manage communications and run the applications.

It provides modularity and clear interfaces, i.e., provides interaction between subsystems.

It ensures the independence between layers by providing the services from lower to higher layer without defining how the services are implemented. Therefore, any modification in a layer will not affect the other layers.

The number of layers, functions, contents of each layer will vary from network to network. However, the purpose of each layer is to provide the service from lower to a higher layer and hiding the details from the layers of how the services are implemented.

The basic elements of layered architecture are services, protocols, and interfaces.

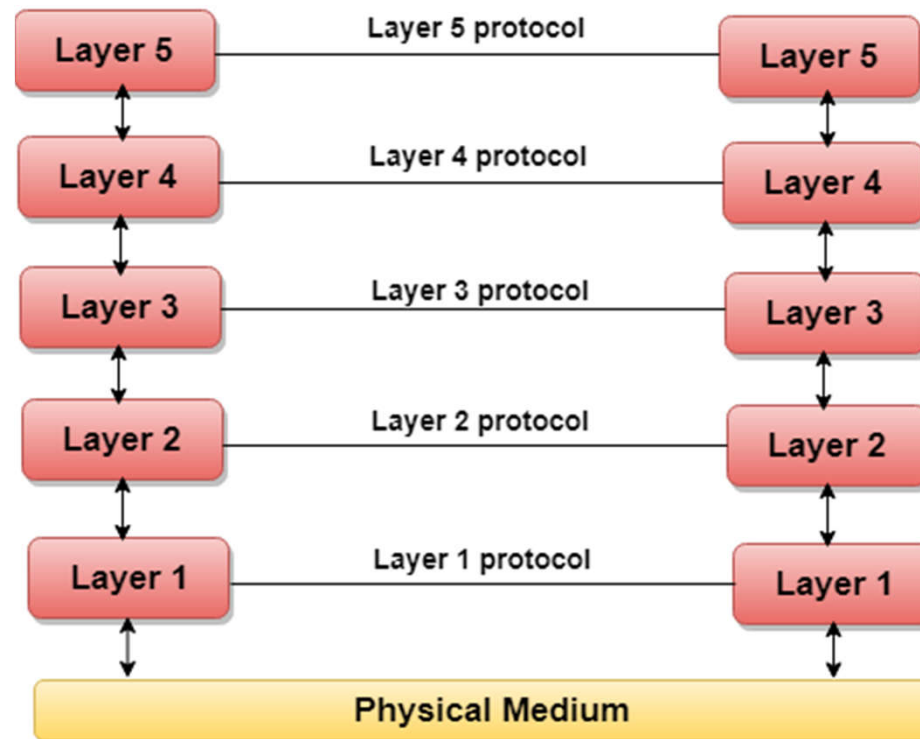
Service: It is a set of actions that a layer provides to the higher layer.

Protocol: It defines a set of rules that a layer uses to exchange the information with peer entity. These rules mainly concern about both the contents and order of the messages used.

Interface: It is a way through which the message is transferred from one layer to another layer.(connection between two layers.)

In a layer n architecture, layer n on one machine will have a communication with the layer n on another machine and the rules used in a conversation are known as a layer-n protocol.

Let's take an example of the five-layered architecture.



In case of layered architecture, no data is transferred from layer n of one machine to layer n of another machine. Instead, each layer passes the data to the layer immediately just below it, until the lowest layer is reached.

Below layer 1 is the physical medium through which the actual communication takes place.

In a layered architecture, unmanageable tasks are divided into several small and manageable tasks.

The data is passed from the upper layer to lower layer through an interface. A Layered architecture provides a clean-cut interface so that minimum information is shared among different layers. It also ensures that the implementation of one layer can be easily replaced by another implementation.

A set of layers and protocols is known as network architecture.

Why do we require Layered architecture?

Divide-and-conquer approach: Divide-and-conquer approach makes a design process in such a way that the unmanageable tasks are divided into small and manageable tasks. In short, we can say that this approach reduces the complexity of the design.

Modularity: Layered architecture is more modular. Modularity provides the independence of layers, which is easier to understand and implement.

Easy to modify: It ensures the independence of layers so that implementation in one layer can be changed without affecting other layers.

Easy to test: Each layer of the layered architecture can be analyzed and tested individually.

OSI Model

OSI stands for **Open System Interconnection** is a reference model that describes how information from a [software](#) application in one [computer](#) moves through a physical medium to the software application in another computer.

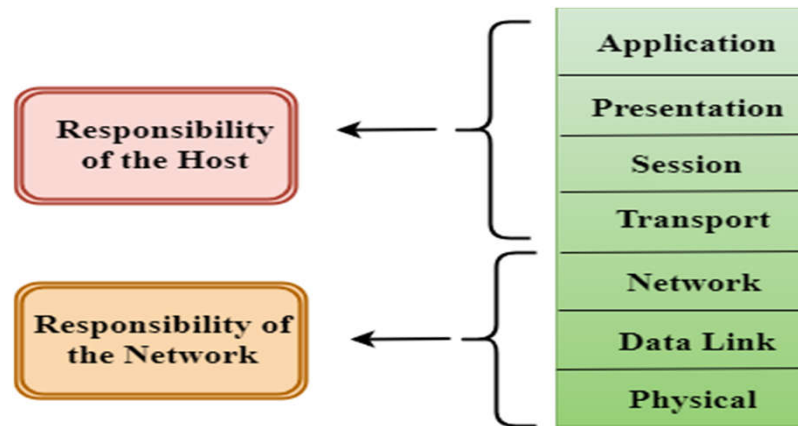
OSI consists of seven layers, and each layer performs a particular network function. OSI model was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications.

OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned a particular task.

Each layer is self-contained, so that task assigned to each layer can be performed independently.

The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software. The OSI model is not a protocol; it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable

Characteristics of OSI Model:



The OSI model is divided into two layers: upper layers and lower layers.

The upper layer of the OSI model mainly deals with the application related issues, and they are implemented only in the software. The application layer is closest to the end user. Both the end user and the application layer interact with the software applications. An upper layer refers to the layer just above another layer.

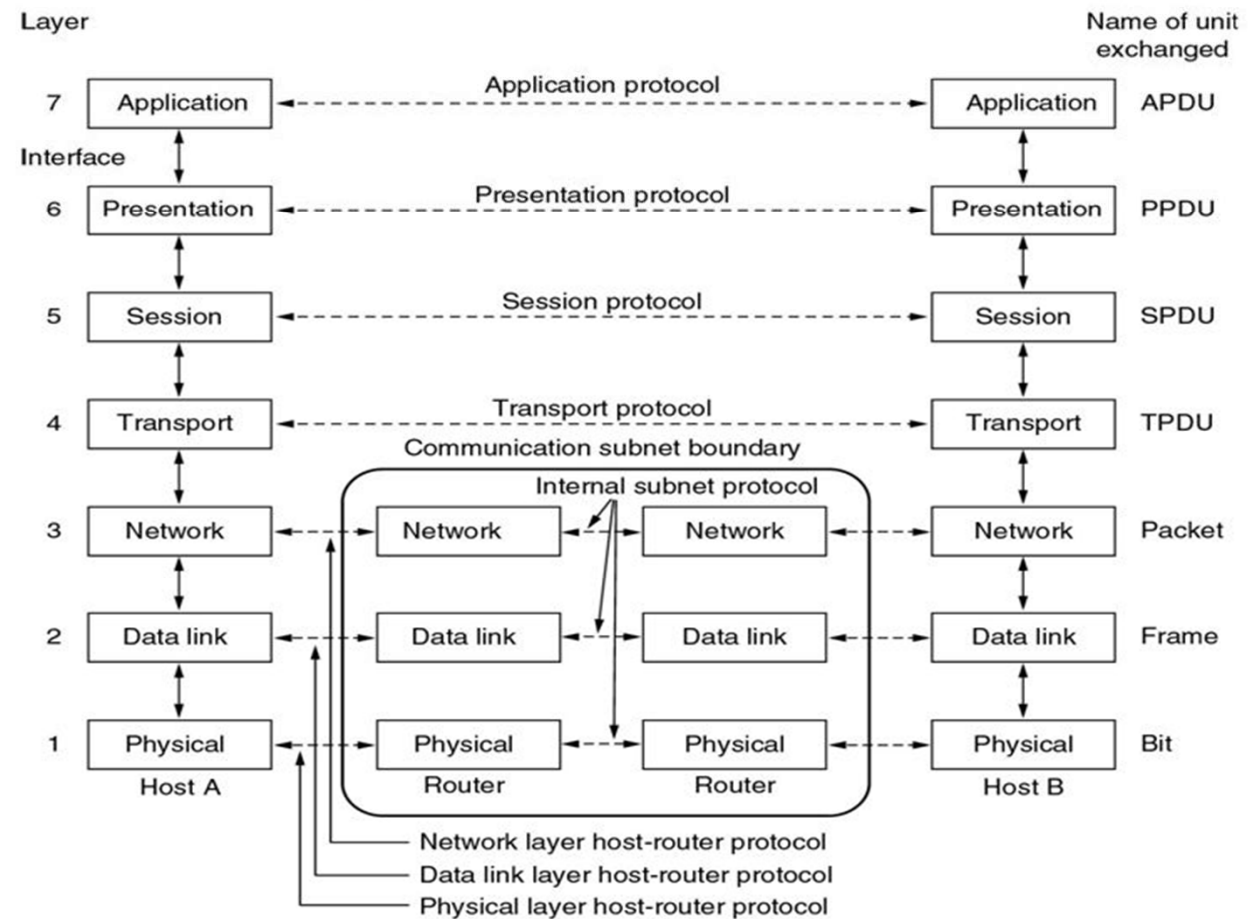
The lower layer of the OSI model deals with the data transport issues. The data link layer and the physical layer are implemented in hardware and software. The physical layer is the lowest layer of the OSI model and is closest to the physical medium. The physical layer is mainly responsible for placing the information on the physical medium.

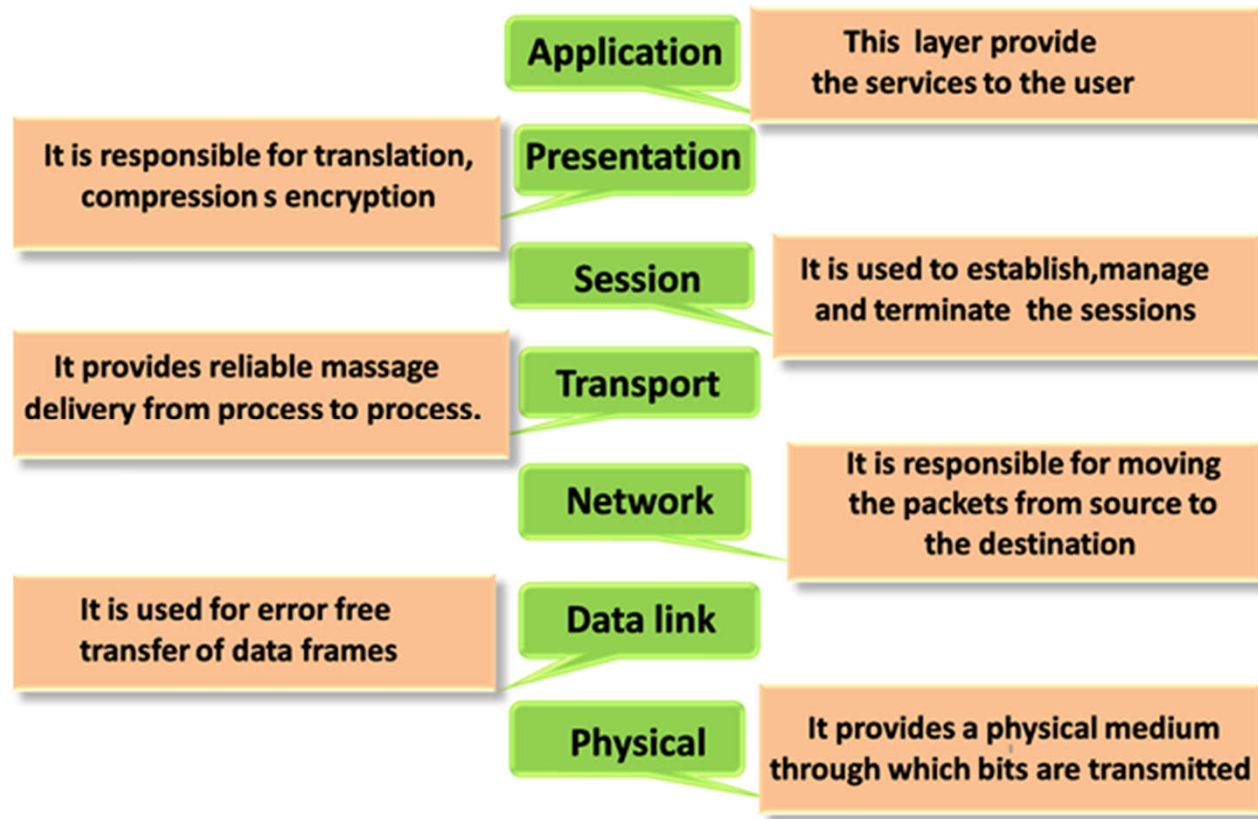
OSI Model

Open System Interconnect is an open standard for all communication systems. OSI model is established by International Standard Organization (ISO). This model has seven layers:

Reference Models

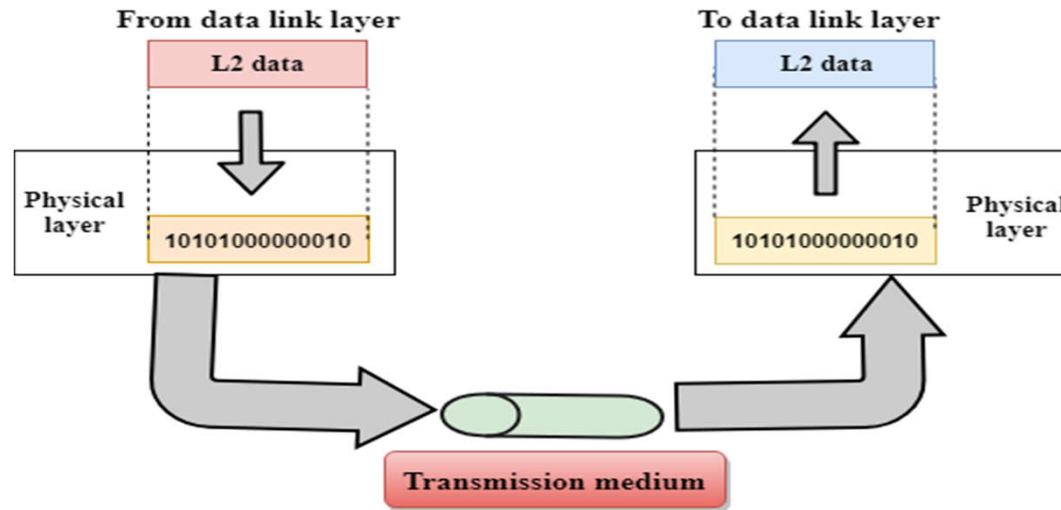
The OSI
reference
model.





Physical layer

- The main functionality of the physical layer is to transmit the individual bits from one node to another node.
- It is the lowest layer of the OSI model.
- It establishes, maintains and deactivates the physical connection.
- It specifies the mechanical, electrical and procedural network interface specifications.



Functions of a Physical layer:

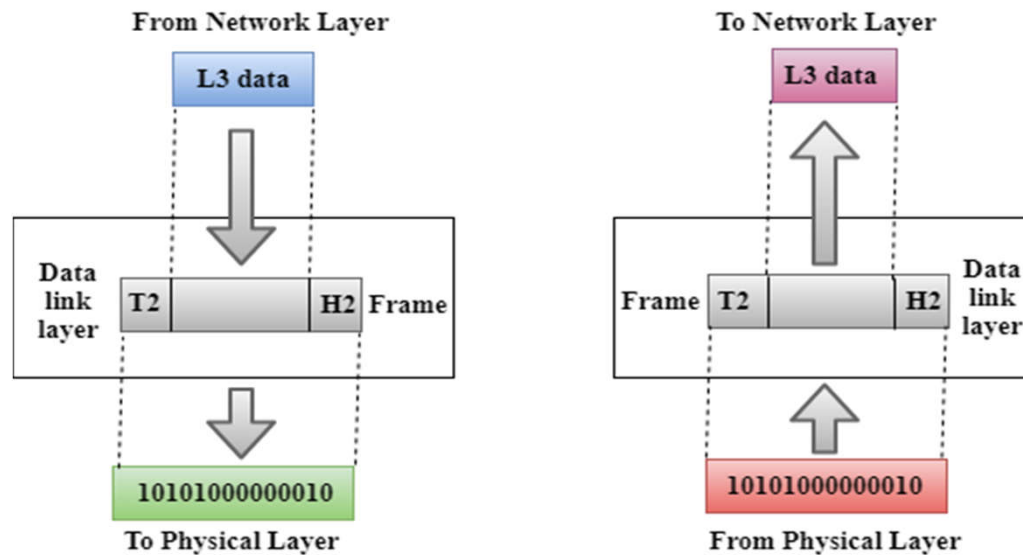
Line Configuration: It defines the way how two or more devices can be connected physically.

Data Transmission: It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.

Topology: It defines the way how network devices are arranged.

Signals: It determines the type of the signal used for transmitting the information. Note::To be transmitted, bits must be encoded into signals--electrical or optical. The physical layer defines the type of encoding (how 0s and 1s are changed to signals).

Data-Link Layer



This layer is responsible for the error-free transfer of data frames.

It defines the format of the data on the network.

It provides a reliable and efficient communication between two or more devices.

It is mainly responsible for the unique identification of each device that resides on a local network.

It contains two sub-layers:

Logical Link Control Layer

It is responsible for transferring the packets to the Network layer of the receiver that is receiving.

It identifies the address of the network layer protocol from the header.

It also provides flow control.

- **Media Access Control Layer**

- A Media access control layer is a link between the Logical Link Control layer and the network's physical layer.
- It is used for transferring the packets over the network.

Functions of the Data-link layer

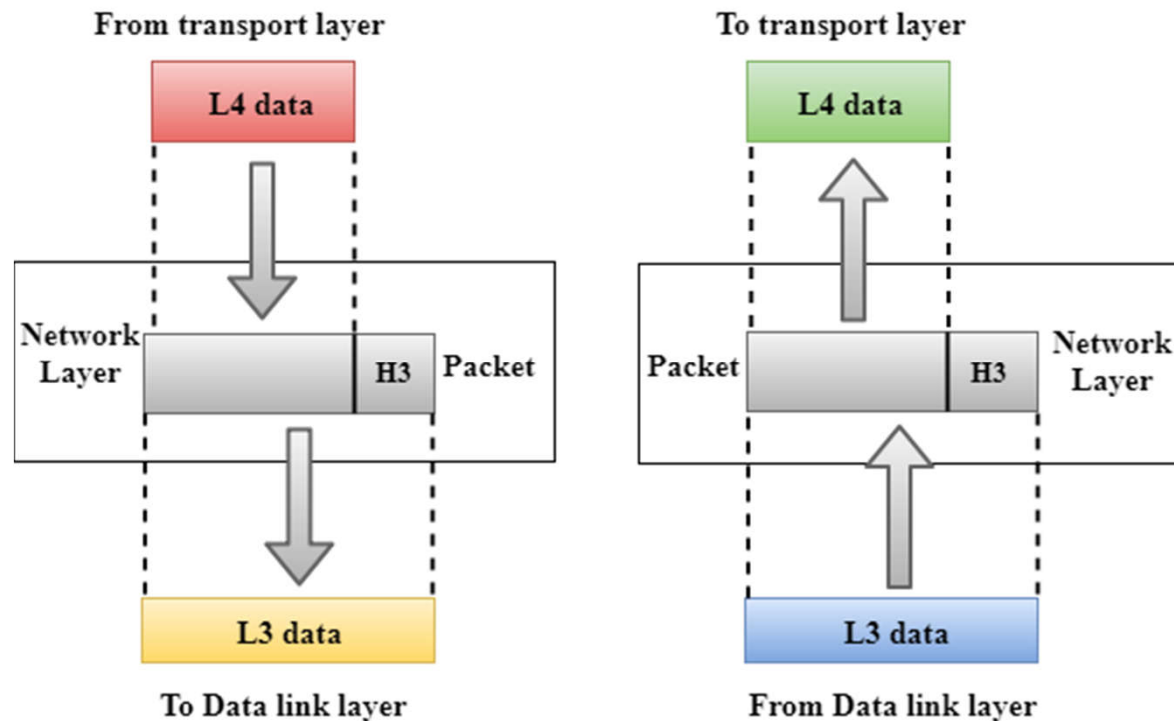
- **Framing:** The data link layer translates the physical's raw bit stream into packets known as Frames. The Data link layer adds the header and trailer to the frame. The header which is added to the frame contains the hardware destination and source address.



Frame Header: It contains the source and the destination addresses of the frame and the control bytes. Payload **field**: It contains the message to be delivered. **Trailer**: It contains the error detection and error correction bits. It is also called a Frame Check Sequence (FCS). Data-link Layer adds also a trailer at the end of each frame. The trailer is responsible for ensuring that frames are received intact or undamaged.

- **Physical Addressing:** The Data link layer adds a header to the frame that contains a destination address. The frame is transmitted to the
 - destination address mentioned in the header.
- **Flow Control:** Flow control is the main functionality of the Data-link layer. It is the technique through which the constant data rate is
 - maintained on both the sides so that no data get corrupted. It ensures that the transmitting station such as a server with higher processing
 - speed does not exceed the receiving station, with lower processing speed.
- **Error Control:** Error control is achieved by adding a calculated value CRC (Cyclic Redundancy Check) that is placed to the Data link layer's trailer
 - which is added to the message frame before it is sent to the physical layer. If any error seems to occur, then the receiver sends the
 - acknowledgment for the retransmission of the corrupted frames.
- **Access Control:** When two or more devices are connected to the same communication channel, then the data link layer protocols
 - are used to determine which device has control over the link at a given time.
- **encapsulation** is used to describe a **process** of adding headers and trailers around some data.

Network Layer



It is a layer 3 that manages device addressing, tracks the location of devices on the network.

It determines the best path to move data from source to the destination based on the network conditions, the priority of service, and other factors.

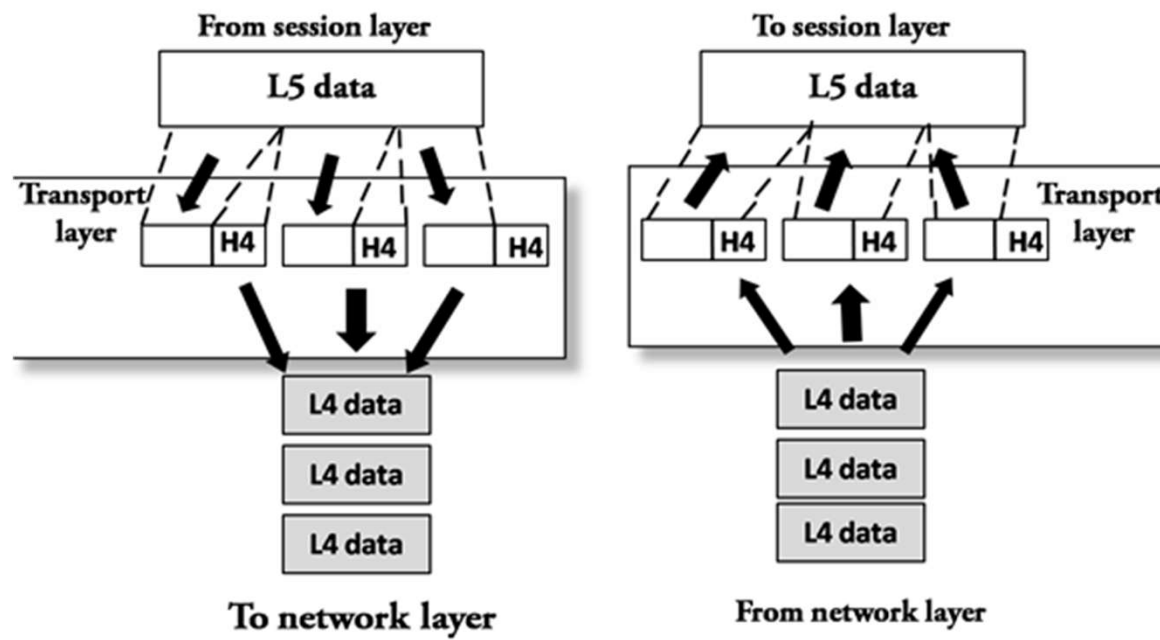
The Data link layer is responsible for routing and forwarding the packets.

Routers are the layer 3 devices, they are specified in this layer and used to provide the routing services within an internetwork.

The protocols used to route the network traffic are known as Network layer protocols. Examples of protocols are IP and Ipv6.

- Functions of Network Layer:
- **Internetworking:** An internetworking is the main responsibility of the network layer. It provides a logical connection between different devices.
- **Addressing**: A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet.
- **Routing**: Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.
- **Packetizing:** A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).

Transport Layer



The Transport layer is a Layer 4 ensures that messages are transmitted in the order in which they are sent and there is no duplication of data.

The main responsibility of the transport layer is to transfer the data completely.

It receives the data from the upper layer and converts them into smaller units known as segments.

This layer can be termed as an end-to-end layer as it provides a point-to-point connection between source and destination to deliver the data reliably.

The two protocols used in this layer are:

Transmission Control Protocol

It is a standard protocol that allows the systems to communicate over the internet.

It establishes and maintains a connection between hosts.

When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments. Each segment travels over the internet using multiple routes, and they arrive in different orders at the destination. The transmission control protocol reorders the packets in the correct order at the receiving end.

User Datagram Protocol

User Datagram Protocol is a transport layer protocol.

It is an unreliable transport protocol as in this case receiver does not send any acknowledgment when the packet is received, the sender does not wait for any acknowledgment. Therefore, this makes a protocol unreliable.

Functions of Transport Layer:

Service-point addressing: Computers run several programs simultaneously due to this reason, the transmission of data from source to the destination not only from one computer to another computer but also from one process to another process. The transport layer adds the header that contains the address known as a service-point address or port address. (The responsibility of the network layer is to transmit the data from one computer to another computer and the responsibility of the transport layer is to transmit the message to the correct process.)

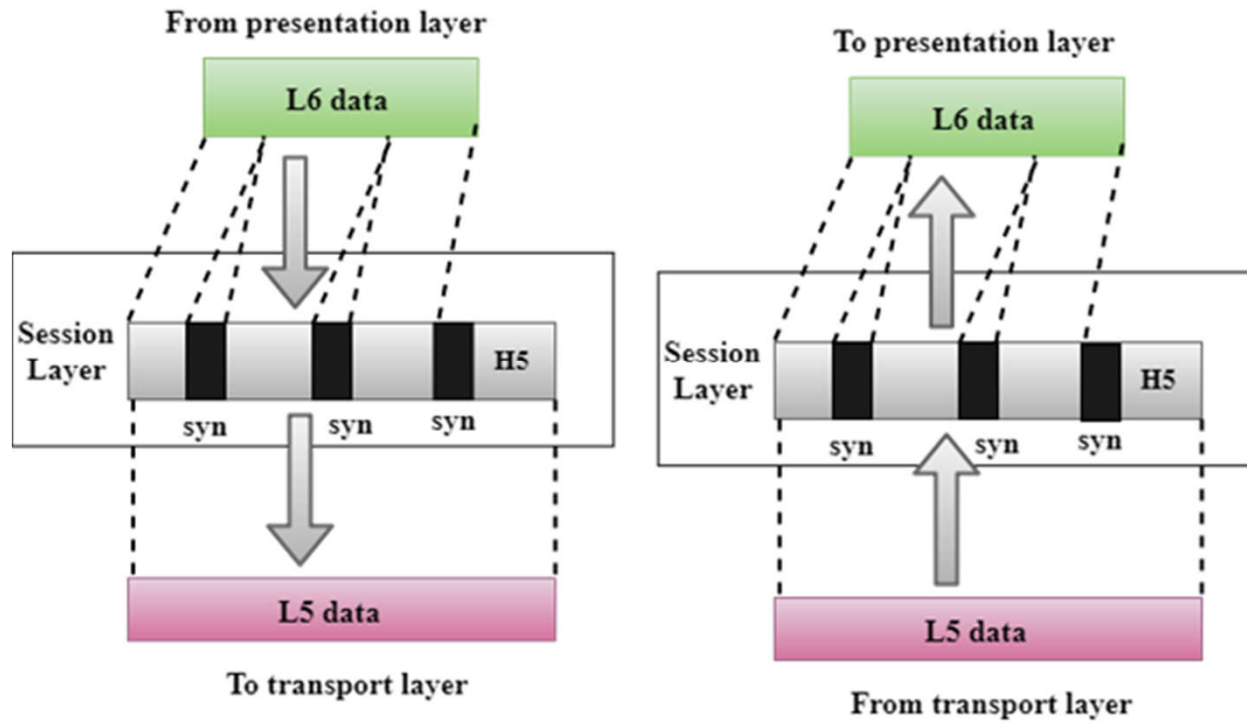
Segmentation and reassembly: When the transport layer receives the message from the upper layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segment. When the message has arrived at the destination, then the transport layer reassembles the message based on their sequence numbers.

Connection control: Transport layer provides two services Connection-oriented service and connectionless service. A connectionless service treats each segment as an individual packet, and they all travel in different routes to reach the destination. A connection-oriented service makes a connection with the transport layer at the destination machine before delivering the packets. In connection-oriented service, all the packets travel in the single route.

Flow control: The transport layer also responsible for flow control but it is performed end-to-end rather than across a single link.

Error control: The transport layer is also responsible for Error control. Error control is performed end-to-end rather than across the single link. The sender transport layer ensures that message reach at the destination without any error.

Session Layer



It is a layer 3 in the OSI model.

The Session layer is used to establish, maintain and synchronizes the interaction between communicating devices.

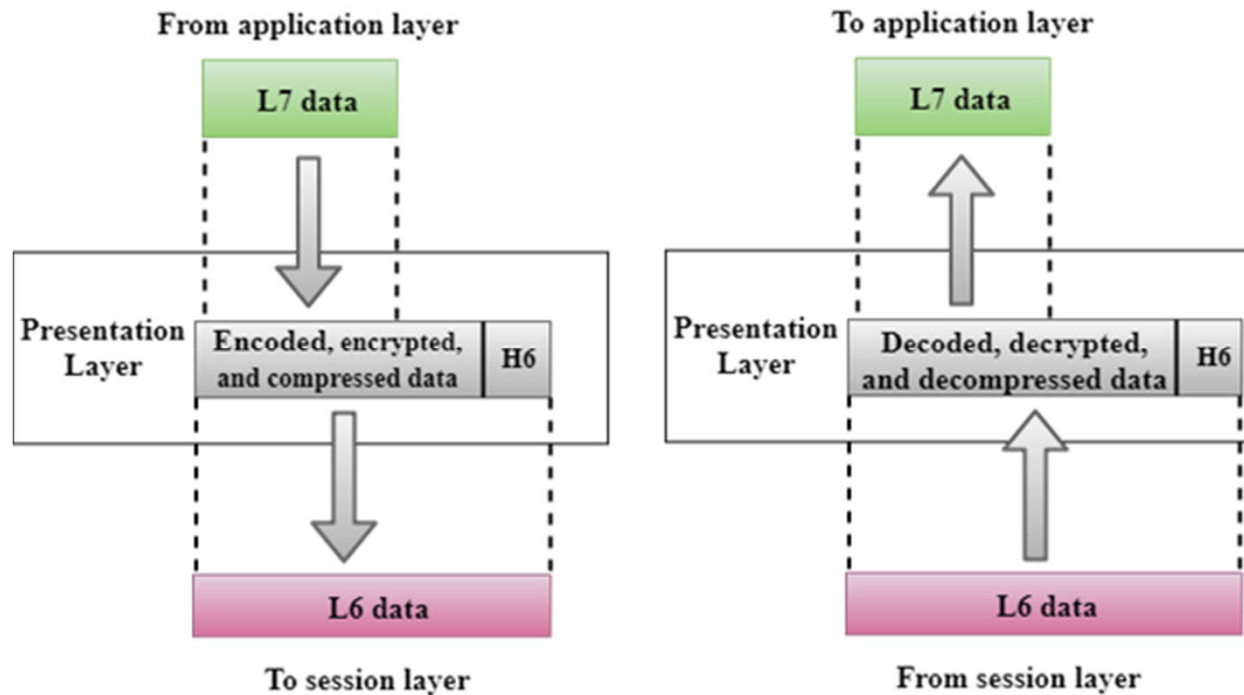
Functions of Session layer:

Dialog control: Session layer acts as a dialog controller that creates a dialog between two processes or we can say that it allows the communication between two processes which can be either half-duplex or full-duplex.

Synchronization: Checkpoints are the midway marks that are added after a particular interval during stream of data transfer. These points are also referred to as synchronization points. Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and recovery.

The Session layer permits process to add these checkpoints. For example, suppose a file of 400 pages is being sent over a network, then it is highly beneficial to set up a checkpoint after every 50 pages so that next 50 pages are sent only when previous pages are received and acknowledged.

Presentation Layer



A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.

It acts as a data translator for a network.

This layer is a part of the operating system that converts the data from one presentation format to another format.

The Presentation layer is also known as the syntax layer.

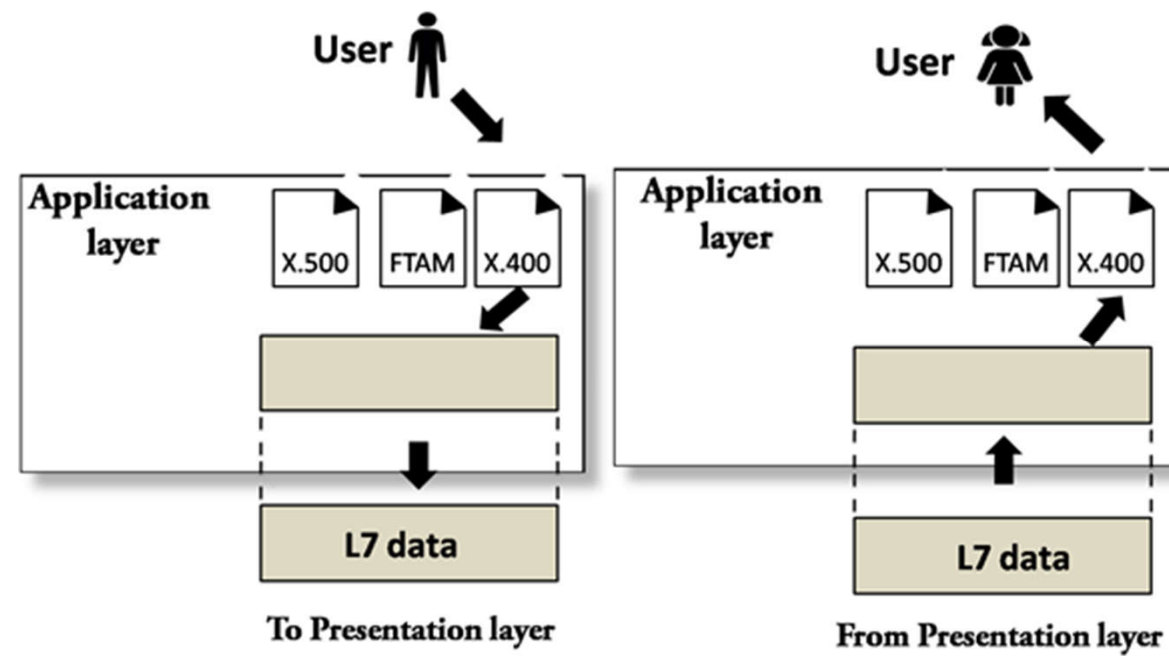
Functions of Presentation layer:

Translation: The processes in two systems exchange the information in the form of character strings, numbers and so on. Different computers use different encoding methods, the presentation layer handles the interoperability between the different encoding methods. It converts the data from sender-dependent format into a common format and changes the common format into receiver-dependent format at the receiving end.

Encryption: Encryption is needed to maintain privacy. Encryption is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.

Compression: Data compression is a process of compressing the data, i.e., it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

Application Layer (Desktop layer-whatever we r seeing on d desktop)



An application layer serves as a window for users and application processes to access network service.

It handles issues such as network transparency, resource allocation, etc.

An application layer is not an application, but it performs the application layer functions.

This layer provides the network services to the end-users.

Functions of Application layer:

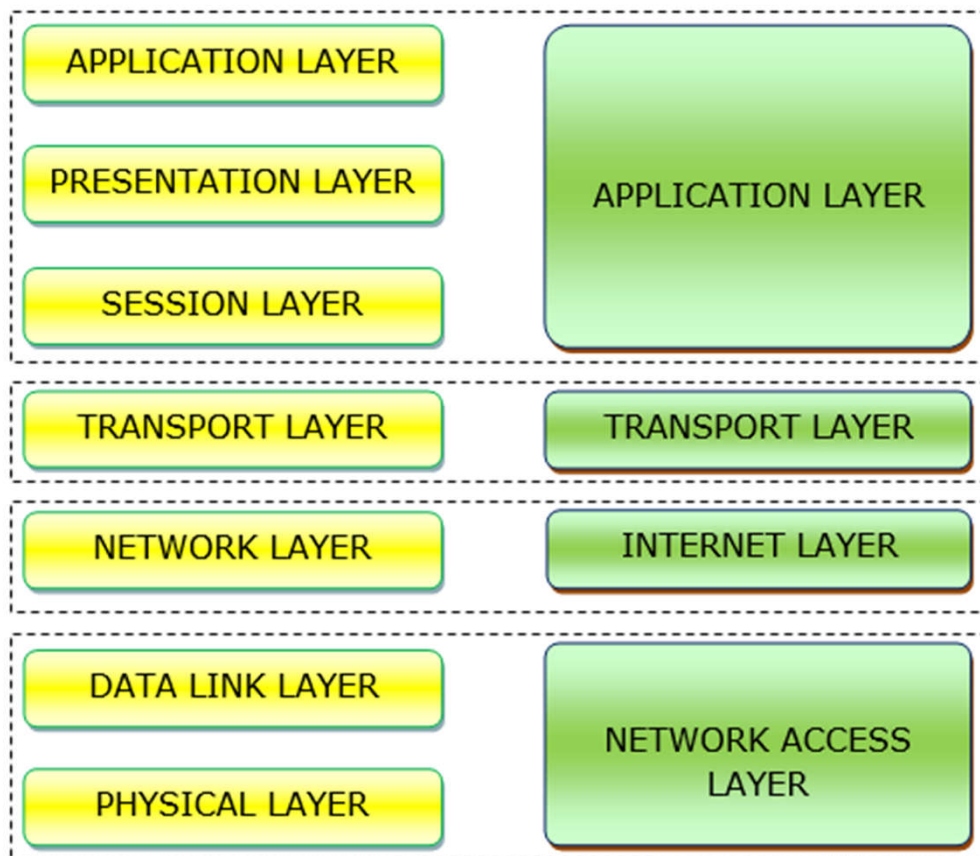
File transfer, access, and management (FTAM): An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.

Mail services: An application layer provides the facility for email forwarding and storage.

Directory services: An application provides the distributed database sources and is used to provide that global information about various objects.

OSI Model

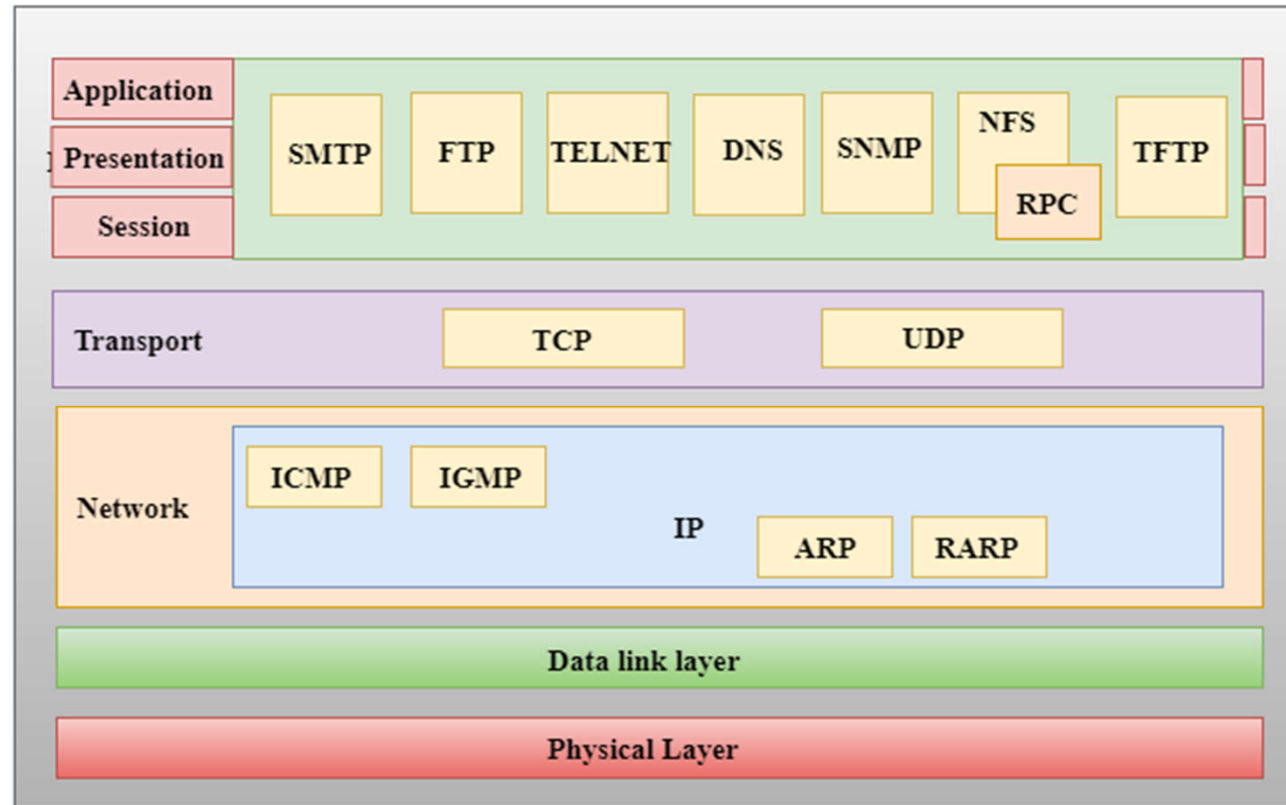
TCP/IP Model



TCP/IP model

- The TCP/IP model was developed prior to the OSI model.
- The TCP/IP model is not exactly similar to the OSI model.
- The TCP/IP model consists of five layers: the application layer, transport layer, network layer, data link layer and physical layer.
- The first three layers provide physical standards, network interface, internetworking, and transport functions that correspond to the first four layers of the OSI model and these four layers are represented in TCP/IP model by a single layer called the application layer.
- TCP/IP is a hierarchical protocol made up of interactive modules, and each of them provides specific functionality.
- Here, hierarchical means that each upper-layer protocol is supported by two or more lower-level protocols.

Functions of TCP/IP layers:



- Network Access Layer
- A network layer is the lowest layer of the TCP/IP model.
- A network layer is the combination of the Physical layer and Data Link layer defined in the OSI reference model.
- It defines how the data should be sent physically through the network.
- This layer is mainly responsible for the transmission of the data between two devices on the same network.
- The functions carried out by this layer are encapsulating the IP datagram into frames transmitted by the network and mapping of IP addresses into physical addresses.
- The protocols used by this layer are ethernet, token ring, FDDI, X.25, frame relay.

Internet Layer

An internet layer is the second layer of the TCP/IP model.

An internet layer is also known as the network layer.

The main responsibility of the internet layer is to send the packets from any network, and they arrive at the destination irrespective of the route they take.

Following are the protocols used in this layer are:

IP Protocol: IP protocol is used in this layer, and it is the most significant part of the entire TCP/IP suite.

Following are the responsibilities of this protocol:

IP Addressing: This protocol implements logical host addresses known as IP addresses. The IP addresses are used by the internet and higher layers to identify the device and to provide internetwork routing.

Host-to-host communication: It determines the path through which the data is to be transmitted.

Data Encapsulation and Formatting: An IP protocol accepts the data from the transport layer protocol. An IP protocol ensures that the data is sent and received securely, it encapsulates the data into message known as IP datagram.

Fragmentation and Reassembly: The limit imposed on the size of the IP datagram by data link layer protocol is known as Maximum Transmission unit (MTU). If the size of IP datagram is greater than the MTU unit, then the IP protocol splits the datagram into smaller units so that they can travel over the local network. Fragmentation can be done by the sender or intermediate router. At the receiver side, all the fragments are reassembled to form an original message.

Routing: When IP datagram is sent over the same local network such as LAN, MAN, WAN, it is known as direct delivery. When source and destination are on the distant network, then the IP datagram is sent indirectly. This can be accomplished by routing the IP datagram through various devices such as routers.

ARP Protocol

ARP stands for **Address Resolution Protocol**.

ARP is a network layer protocol which is used to find the physical address from the IP address.

The two terms are mainly associated with the ARP Protocol:

ARP request: When a sender wants to know the physical address of the device, it broadcasts the ARP request to the network.

ARP reply: Every device attached to the network will accept the ARP request and process the request, but only recipient recognize the IP address and sends back its physical address in the form of ARP reply. The recipient adds the physical address both to its cache memory and to the datagram header

RARP

RARP is a network layer protocol which is used to find the IP address from physical address.

ICMP Protocol

ICMP stands for Internet Control Message Protocol.

It is a mechanism used by the hosts or routers to send notifications regarding datagram problems back to the sender.

A datagram travels from router-to-router until it reaches its destination. If a router is unable to route the data because of some unusual conditions such as disabled links, a device is on fire or network congestion, then the ICMP protocol is used to inform the sender that the datagram is undeliverable.

An ICMP protocol mainly uses two terms:

ICMP Test: ICMP Test is used to test whether the destination is reachable or not.

ICMP Reply: ICMP Reply is used to check whether the destination device is responding or not.

The core responsibility of the ICMP protocol is to report the problems, not correct them. The responsibility of the correction lies with the sender.

ICMP can send the messages only to the source, but not to the intermediate routers because the IP datagram carries the addresses of the source and destination but not of the router that it is passed to.

Transport Layer

The transport layer is responsible for the reliability, flow control, and correction of data which is being sent over the network.

The two protocols used in the transport layer are **User Datagram protocol and Transmission control protocol.**

•User Datagram Protocol (UDP)

- It provides connectionless service and end-to-end delivery of transmission.
- It is an unreliable protocol as it discovers the errors but not specify the error.
- User Datagram Protocol discovers the error, and ICMP protocol reports the error to the sender that user datagram has been damaged.

•UDP consists of the following fields:

Source port address: The source port address is the address of the application program that has created the message.

Destination port address: The destination port address is the address of the application program that receives the message.

Total length: It defines the total number of bytes of the user datagram in bytes.

Checksum: The checksum is a 16-bit field used in error detection.

- UDP does not specify which packet is lost. UDP contains only checksum; it does not contain any ID of a data segment.



Header Format

Source port address 16 bits	Destination port address 16 bits
Total length 16 bits	Checksum 16 bits

- Transmission Control Protocol (TCP)**

- It provides a full transport layer services to applications.
- It creates a virtual circuit between the sender and receiver, and it is active for the duration of the transmission.
- TCP is a reliable protocol as it detects the error and retransmits the damaged frames. Therefore, it ensures all the segments must be received and acknowledged before the transmission is considered to be completed and a virtual circuit is discarded.
- At the sending end, TCP divides the whole message into smaller units known as segment, and each segment contains a sequence number which is required for reordering the frames to form an original message.
- At the receiving end, TCP collects all the segments and reorders them based on sequence numbers.

Application Layer

An application layer is the topmost layer in the TCP/IP model.

It is responsible for handling high-level protocols, issues of representation.

This layer allows the user to interact with the application.

When one application layer protocol wants to communicate with another application layer, it forwards its data to the transport layer.

There is an ambiguity occurs in the application layer. Every application cannot be placed inside the application layer except those who interact with the communication system. For example: text editor cannot be considered in application layer while web browser using **HTTP** protocol to interact with the network where **HTTP** protocol is an application layer protocol.

Following are the main protocols used in the application layer:

HTTP: HTTP stands for Hypertext transfer protocol. This protocol allows us to access the data over the world wide web. It transfers the data in the form of plain text, audio, video. It is known as a Hypertext transfer protocol as it has the efficiency to use in a hypertext environment where there are rapid jumps from one document to another.

SNMP: SNMP stands for Simple Network Management Protocol. It is a framework used for managing the devices on the internet by using the TCP/IP protocol suite.

SMTP: SMTP stands for Simple mail transfer protocol. The TCP/IP protocol that supports the e-mail is known as a Simple mail transfer protocol. This protocol is used to send the data to another e-mail address.

DNS: DNS stands for Domain Name System(The **Domain Name System (DNS)** is the Internet's **system** for mapping alphabetic **names** to numeric Internet Protocol (IP) addresses). An IP address is used to identify the connection of a host to the internet uniquely. But, people prefer to use the names instead of addresses. Therefore, the system that maps the name to the address is known as Domain Name System.

TELNET: It is an abbreviation for Terminal Network. It establishes the connection between the local computer and remote computer in such a way that the local terminal appears to be a terminal at the remote system.

FTP: FTP stands for File Transfer Protocol. FTP is a standard internet protocol used for transmitting the files from one computer to another computer.

Next Topic

Similarities between OSI and TCP / IP Reference Models

- Both the reference models are based upon layered architecture.
- The layers in the models are compared with each other. The physical layer and the data link layer of the OSI model correspond to the link layer of the TCP/IP model. The network layers and the transport layers are the same in both the models. The session layer, the presentation layer and the application layer of the OSI model together form the application layer of the TCP/IP model.
- In both the models, protocols are defined in a layer-wise manner.
- In both models, data is divided into packets and each packet may take the individual route from the source to the destination.

Differences between OSI and TCP / IP Reference Models

- OSI model is a generic model that is based upon functionalities of each layer. TCP/IP model is a protocol-oriented standard.
- OSI model distinguishes the three concepts, namely, services, interfaces, and protocols. TCP/IP does not have a clear distinction between these three.
- OSI model gives guidelines on how communication needs to be done, while TCP/IP protocols layout standards on which the Internet was developed. So, TCP/IP is a more practical model.
- In OSI, the model was developed first and then the protocols in each layer were developed. In the TCP/IP suite, the protocols were developed first and then the model was developed.
- The OSI has seven layers while the TCP/IP has four layers.

Physical Layer Introduction

- Physical layer in the OSI model plays the role of interacting with actual hardware and signaling mechanism. Physical layer is the only layer of OSI network model which actually deals with the physical connectivity of two different stations. This layer defines the hardware equipment, cabling, wiring, frequencies, pulses used to represent binary signals etc.
- Physical layer provides its services to Data-link layer. Data-link layer hands over frames to physical layer. Physical layer converts them to electrical pulses, which represent binary data. The binary data is then sent over the wired or wireless media.

- Signals
- When data is sent over physical medium, it needs to be first converted into electromagnetic signals. Data itself can be analog such as human voice, or digital such as file on the disk. Both analog and digital data can be represented in digital or analog signals.
- **Digital Signals**
- Digital signals are discrete in nature and represent sequence of voltage pulses. Digital signals are used within the circuitry of a computer system.
- **Analog Signals**
- Analog signals are in continuous wave form in nature and represented by continuous electromagnetic waves.
- Transmission Impairment
- When signals travel through the medium they tend to deteriorate. This may have many reasons as given:
- **Attenuation**
- For the receiver to interpret the data accurately, the signal must be sufficiently strong. When the signal passes through the medium, it tends to get weaker. As it covers distance, it loses strength.
- **Dispersion**
- As signal travels through the media, it tends to spread and overlaps. The amount of dispersion depends upon the frequency used.
- **Delay distortion**
- Signals are sent over media with pre-defined speed and frequency. If the signal speed and frequency do not match, there are possibilities that signal reaches destination in arbitrary fashion. In digital media, this is very critical that some bits reach earlier than the previously sent ones.
-

- Transmission Media
- The media over which the information between two computer systems is sent, called transmission media. Transmission media comes in two forms.
- **Guided Media**
- All communication wires/cables are guided media, such as UTP, coaxial cables, and fiber Optics. In this media, the sender and receiver are directly connected and the information is send (guided) through it.
- **Unguided Media**
- Wireless or open air space is said to be unguided media, because there is no connectivity between the sender and receiver. Information is spread over the air, and anyone including the actual recipient may collect the information.
- Channel Capacity
- The speed of transmission of information is said to be the channel capacity. We count it as data rate in digital world. It depends on numerous factors such as:
- **Bandwidth:** The physical limitation of underlying media.
- **Error-rate:** Incorrect reception of information because of noise.
- **Encoding:** The number of levels used for signaling.
- **(encoding or modulation techniques for transforming digital data into analog signals.**
- We have **used two voltage levels** to send a single bit over a single **signal**. But if we **use** more than **3 voltage levels**, we can send more bits per **signal**. For example, if **6 voltage levels** are **used** to represent 8 bits on a single **signal**, then such **encoding** is termed as **8B/6T encoding**.)

- Write Details
- COMPUTER NETWORK MODELS

- A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.
- A **computer** is a machine that can be programmed to carry out sequences of arithmetic or logical operations automatically. Modern **computers** can perform generic sets of operations known as programs.

Basics of Computer Networking

- **Open system:**
A system which is connected to the network and is ready for communication.
- **Closed system:**
A system which is not connected to the network and can't be communicated with.
- **Computer Network:**
An interconnection of multiple devices, also known as hosts, that are connected using multiple paths for the purpose of sending/receiving data or media. Computer networks can also include multiple devices/mediums which help in the communication between two different devices; these are known as **Network devices** and include things such as routers, switches, hubs, and bridges.
-



Router



Hub



Bridge



Wireless
Router



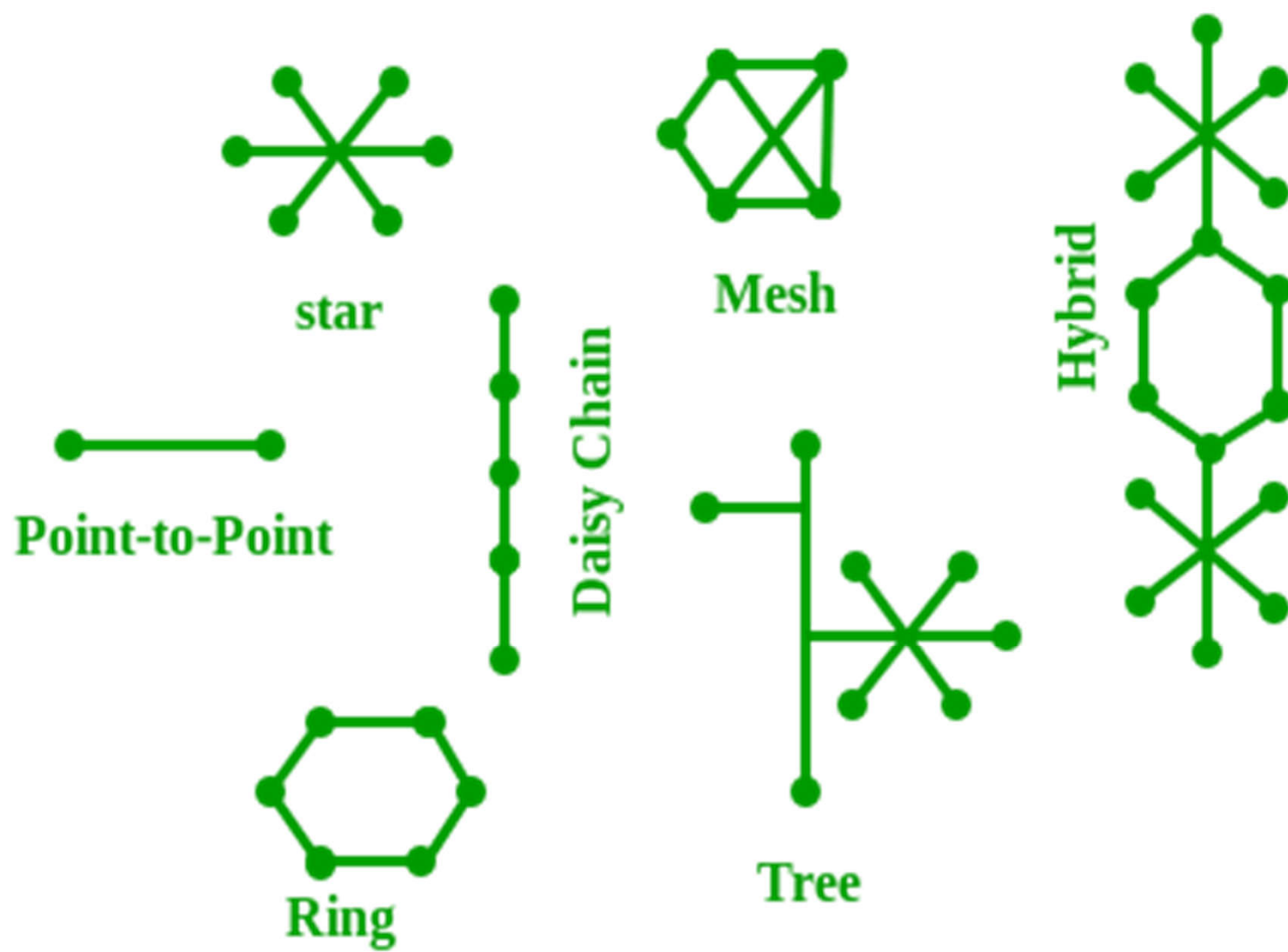
Switch



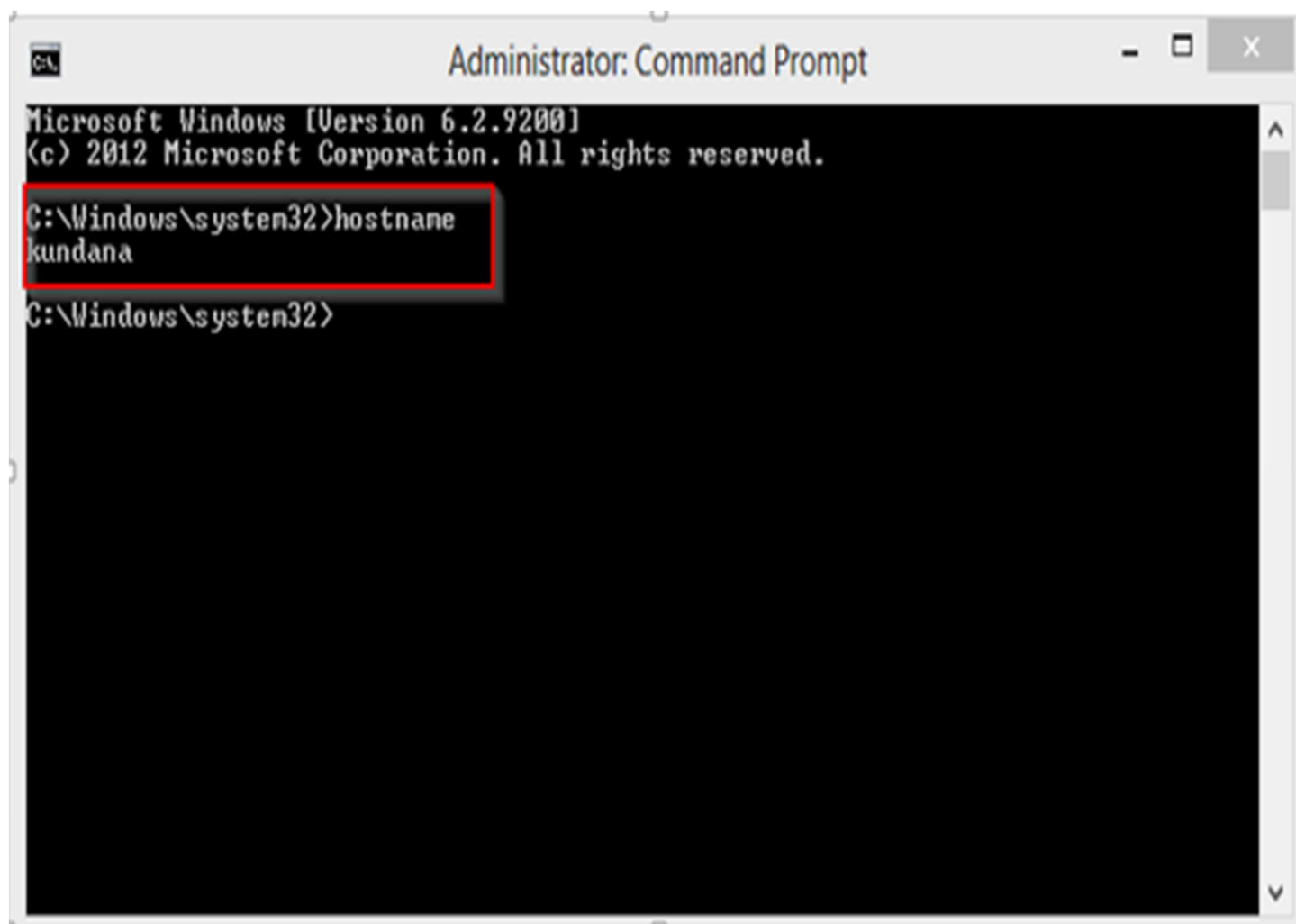
Wireless
Bridge

- **Network Topology:**

The layout arrangement of the different devices in a network. Common examples include: Bus, Star, Mesh, Ring, and Daisy chain.



- **OSI:**
OSI stands for **Open Systems Interconnection**. It is a reference model that specifies standards for communications protocols and also the functionalities of each layer.
- **Protocol:**
A protocol is the set of rules or algorithms which define the way how two entities can communicate across the network and there exists different protocol defined at each layer of the OSI model. Few of such protocols are TCP, IP, UDP, ARP, DHCP, FTP and so on.
- **UNIQUE IDENTIFIERS OF NETWORK**
Host name:
Each device in the network is associated with a unique device name known as Hostname.
Type “hostname” in the command prompt(Administrator Mode) and press ‘Enter’, this displays the hostname of your machine.



Administrator: Command Prompt

```
Microsoft Windows [Version 6.2.9200]  
(c) 2012 Microsoft Corporation. All rights reserved.  
C:\Windows\system32>hostname  
kundana  
C:\Windows\system32>
```

- **IP Address (Internet Protocol address):**

Also known as the Logical Address, the IP Address is the network address of the system across the network.

To identify each device in the world-wide-web, the Internet Assigned Numbers Authority (IANA) assigns an IPV4 (Version 4) address as a unique identifier to each device on the Internet.

The length of an IPv4 address is 32-bits, hence, we have 2^{32} IP addresses available. The length of an IPv6 address is 128-bits.

Type "ipconfig" in the command prompt and press 'Enter', this gives us the IP address of the device.

- **MAC Address (Media Access Control address):**

Also known as physical address, the MAC Address is the unique identifier of each host and is associated with its NIC (Network Interface Card). Network Interface Card (NIC) is a **hardware component** that is present on the computer. It is used to **connect different networking devices** such as computers and servers to share data over the connected network.

A MAC address is assigned to the NIC at the time of manufacturing.

The length of the MAC address is : 12-nibble/ 6 bytes/ 48 bits

Type "ipconfig/all" in the command prompt and press 'Enter', this gives us the MAC address.

- **Port:**
A port can be referred to as a logical channel through which data can be sent/received to an application. Any host may have multiple applications running, and each of these applications is identified using the port number on which they are running.
A port number is a 16-bit integer, hence, we have 2^{16} ports available which are categorized as shown below:
Port Types
 - Range
 - Well known Ports 0 – 1023
 - Registered Ports 1024 – 49151
 - Ephemeral Ports 49152 – 65535
 - Number of ports: 65,536
Range: 0 – 65535
*Type “**netstat -a**” in the command prompt and press ‘Enter’, this lists all the ports being used.*

Administrator: Command Prompt

```
C:\Windows\system32>netstat -a
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	kundana:0	LISTENING
TCP	0.0.0.0:443	kundana:0	LISTENING
TCP	0.0.0.0:445	kundana:0	LISTENING
TCP	0.0.0.0:902	kundana:0	LISTENING
TCP	0.0.0.0:912	kundana:0	LISTENING
TCP	0.0.0.0:8092	kundana:0	LISTENING
TCP	0.0.0.0:49152	kundana:0	LISTENING
TCP	0.0.0.0:49153	kundana:0	LISTENING
TCP	0.0.0.0:49154	kundana:0	LISTENING
TCP	0.0.0.0:49155	kundana:0	LISTENING
TCP	0.0.0.0:49158	kundana:0	LISTENING
TCP	0.0.0.0:49160	kundana:0	LISTENING
TCP	127.0.0.1:5354	kundana:0	LISTENING
TCP	127.0.0.1:5939	kundana:0	LISTENING
TCP	127.0.0.1:8307	kundana:0	LISTENING
TCP	127.0.0.1:30000	kundana:0	LISTENING
TCP	192.168.1.99:139	kundana:0	LISTENING

- **Socket:**

The unique combination of IP address and Port number together are termed as Socket.

- **Other related concepts**

DNS Server:

DNS stands for **Domain Name system**.

DNS is basically a server which translates web addresses or URLs (ex: www.google.com) into their corresponding IP addresses. We don't have to remember all the IP addresses of each and every website.

The command '**nslookup**' gives you the IP address of the domain you are looking for. This also provides the information of our DNS Server.

C:\

Administrator: Command Prompt

```
C:\Windows\system32>nslookup www.google.com
```

```
Server: UnKnown
```

```
Address: 192.168.43.1
```

```
Non-authoritative answer:
```

```
Name: www.google.com
```

```
Addresses: 2404:6800:4003:c00::6a
```

```
74.125.200.106
```

```
74.125.200.104
```

```
74.125.200.99
```

```
74.125.200.103
```

```
74.125.200.147
```

```
74.125.200.105
```

```
C:\Windows\system32>
```

- **ARP:**

ARP stands for **Address Resolution Protocol**.

It is used to convert an IP address to its corresponding physical address(i.e., MAC Address).

ARP is used by the Data Link Layer to identify the MAC address of the Receiver's machine.

- **RARP:**

RARP stands for **Reverse Address Resolution Protocol**.

As the name suggests, it provides the IP address of the device given a physical address as input. But RARP has become obsolete since the time DHCP has come into the picture.